

Indoor Gardening Services Management (IGSM)

Final Project Report

Project	Indoor Gardening Services Management (IGSM)
Done By	Dega Leela Ramana
Platform	ServiceNow Scoped Application
Project Type	Enterprise Workflow Automation Application
GitHub Repository	github.com/Dega-Leela-Ramana/Indoor-Gardening-Service-management

Table of contents

1. Introduction
2. Business objective
3. Requirement analysis
4. System design
5. Implementation
6. Project planning and scheduling
7. Functional and performance testing
8. Results and discussion
9. Advantages and limitations
10. Conclusion
11. Future scope
12. Appendix

1. Introduction

The Indoor Gardening Services Management (IGSM) application was developed as a complete ServiceNow scoped application to automate indoor gardening service operations. The system replaces manual request handling with a structured digital workflow that manages service requests, technician assignments, approvals, SLA monitoring, and request tracking. The project was implemented using ServiceNow Studio, custom tables, Service Catalog, Flow Designer, Access Control Lists (ACLs), Business Rules, and Client Scripts. The application allows users to submit indoor gardening service requests for plant installation, maintenance, watering, pruning, replacement, and inspection. Once a request is submitted, the workflow automatically routes it for approval, assigns it to the appropriate technician group, tracks SLA performance, and updates the request state until completion. The solution improves operational efficiency, transparency, accountability, and customer satisfaction by ensuring every request follows a standardized lifecycle.

2. Business objective

The primary objective of IGSM is to create a centralized service management platform for indoor gardening operations. The application was designed to eliminate delays caused by manual communication, improve technician coordination, and provide real-time visibility into request status and service performance. The business goals achieved through this project include automated request creation, approval management, technician assignment, SLA monitoring, role-based access control, and reporting. Managers can monitor pending requests and SLA compliance, while technicians receive assigned work items through automated workflows. The solution demonstrates how ServiceNow can be used beyond traditional IT service management for operational workflow automation.

3. Requirement analysis

The requirement analysis phase focused on identifying the functional and security requirements of the indoor gardening workflow. The application required a custom request table, Service Catalog integration, user roles, technician groups, approval processing, and SLA monitoring. A custom table named Indoor Garden Request was created to store gardening requests. User groups were configured for Indoor Gardening Technicians and Indoor Gardening Managers. Access Control Lists (ACLs) were implemented to ensure that requesters could access only their own requests while technicians and managers had appropriate operational permissions. The following screenshots document the implemented data model, user groups, and security configuration.

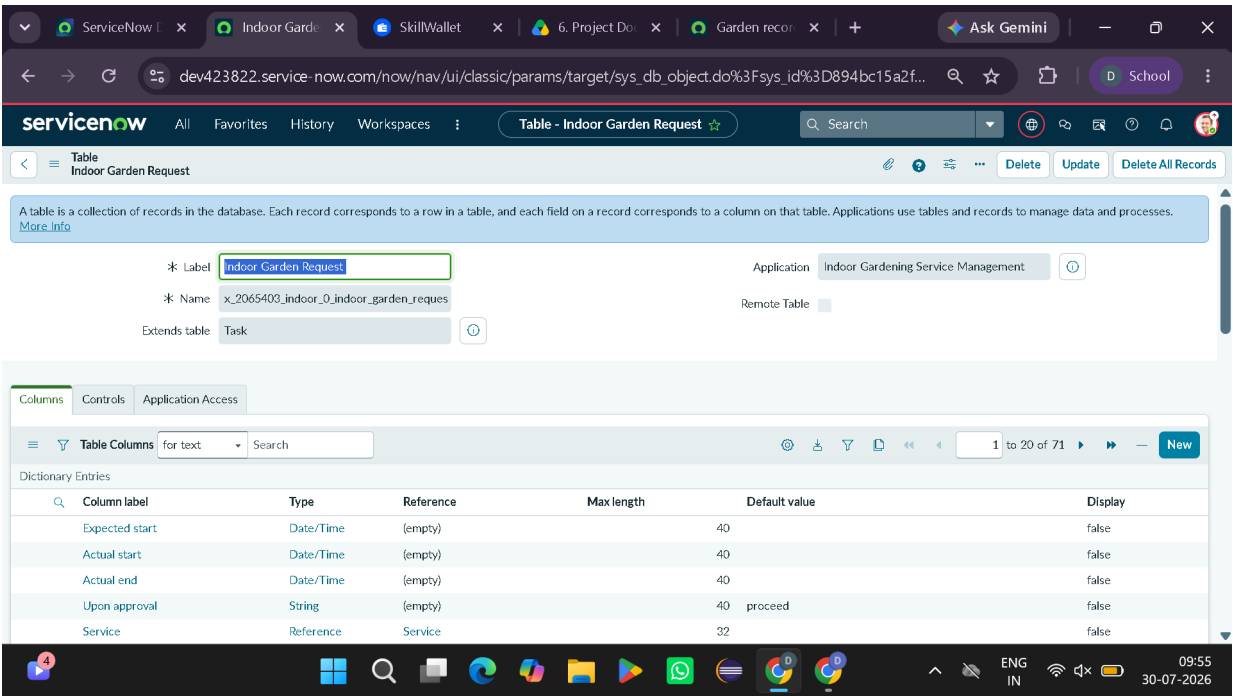


Figure: Indoor Garden Request custom table created in the IGSM scoped application.

3.1 Record producer and customer request interface

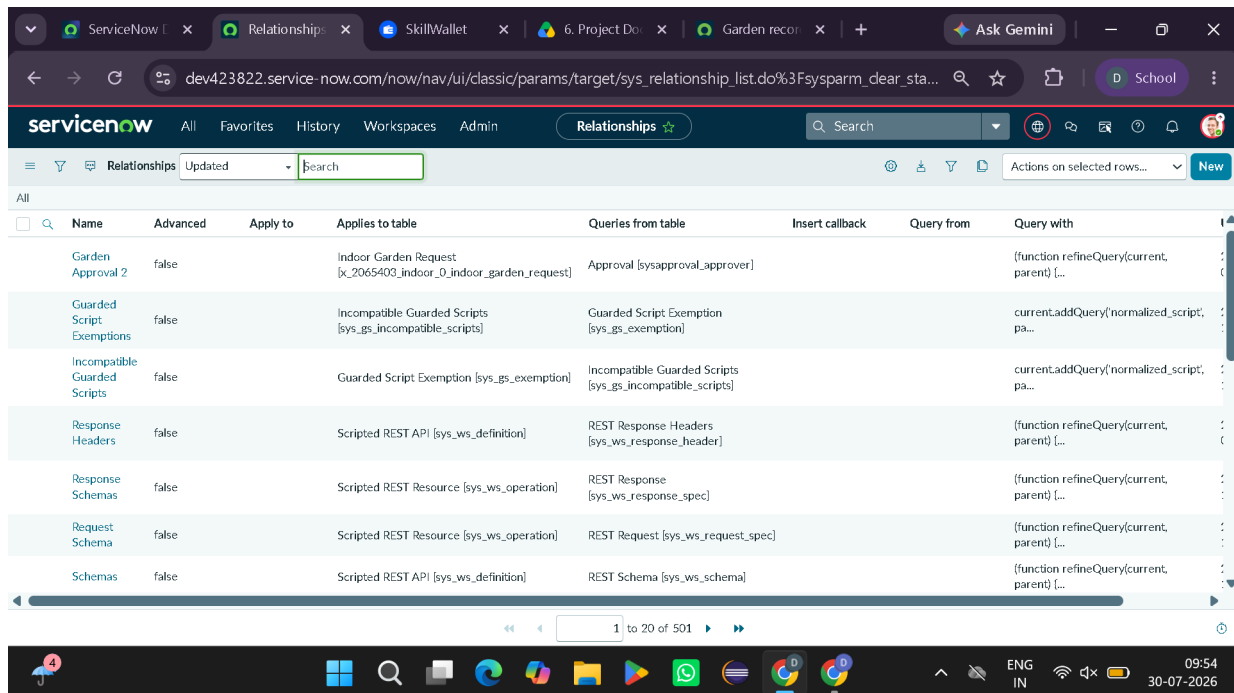
The Garden Record Producer was created within the Service Catalog to provide a user-friendly request submission interface. Users can enter the requested plant name, location, plant type, service type, preferred date, priority, and additional description. The Record Producer maps the submitted variables directly to the Indoor Garden Request table, ensuring consistent data creation. This implementation demonstrates the customer-facing component of the IGSM application and forms the starting point of the automated request workflow.

The screenshot displays the ServiceNow web interface for the 'Garden record producer'. The browser's address bar shows the URL: `dev423822.service-now.com/sp?id=sc_cat_item&sys_id=209f75e2029f489d84a2e9e33f18e608`. The ServiceNow header includes the logo and navigation links: Knowledge, Catalog, Requests, System Status, Cart, Tours, and a user profile for 'System Administrator'. The breadcrumb trail indicates the path: Home > Service Catalog > Office > Services > Garden record producer. A search bar labeled 'Search Catalog' is present. The main form area is titled 'Garden record producer' with the subtitle 'Submit an indoor gardening service request'. It contains several required fields marked with an asterisk: 'What is the requested name.?' (a text input field), 'Where is the location.?' (a text input field), 'What is plant type' (a radio button group with options: Flowering Plants, Foliage Plants, Succulents, Herbs), and 'What service type u need' (a radio button group with options: New Installation, Maintenance). To the right of the form, there is a 'Save as Draft' button and a 'Submit' button. Below these buttons, a 'Required information' section lists the fields: 'What is the requested name.?', 'Where is the location.?', and 'Preferred date'. The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 09:48 on 30-07-2026.

Figure: Garden Record Producer used for customer request submission.

4. System design

The system architecture consists of the Record Producer, Indoor Garden Request table, Flow Designer workflow, approval processing, technician assignment, SLA engine, ACL security, and notification mechanisms. The request lifecycle follows New → Pending Approval → Assigned → In Progress → Completed → Closed. An approval relationship was configured to connect gardening requests with the ServiceNow approval engine. This ensures that manager approvals can be tracked and audited within the request lifecycle. The architecture separates customer interaction, workflow automation, security, and SLA monitoring into modular components, improving maintainability and scalability.



The screenshot displays the ServiceNow 'Relationships' table. The table lists various system components and their associated queries and scripts. The first row, 'Garden Approval 2', is highlighted, showing its relationship to the 'Indoor Garden Request' table and the 'Approval' query. Other rows include 'Guarded Script Exemptions', 'Incompatible Guarded Scripts', 'Response Headers', 'Response Schemas', 'Request Schema', and 'Schemas'.

Name	Advanced	Apply to	Applies to table	Queries from table	Insert callback	Query from	Query with
Garden Approval 2	false		Indoor Garden Request [x_2065403_indoor_0_indoor_garden_request]	Approval [sysapproval_approver]			{function refineQuery(current, parent) [...]
Guarded Script Exemptions	false		Incompatible Guarded Scripts [sys_gs_incompatible_scripts]	Guarded Script Exemption [sys_gs_exemption]			current.addQuery('normalized_script', pa...
Incompatible Guarded Scripts	false		Guarded Script Exemption [sys_gs_exemption]	Incompatible Guarded Scripts [sys_gs_incompatible_scripts]			current.addQuery('normalized_script', pa...
Response Headers	false		Scripted REST API [sys_ws_definition]	REST Response Headers [sys_ws_response_header]			{function refineQuery(current, parent) [...]
Response Schemas	false		Scripted REST Resource [sys_ws_operation]	REST Response [sys_ws_response_spec]			{function refineQuery(current, parent) [...]
Request Schema	false		Scripted REST Resource [sys_ws_operation]	REST Request [sys_ws_request_spec]			{function refineQuery(current, parent) [...]
Schemas	false		Scripted REST API [sys_ws_definition]	REST Schema [sys_ws_schema]			{function refineQuery(current, parent) [...]

Figure: Approval relationship configured for the Indoor Garden Request workflow.

4.1 ServiceNow Studio application architecture

The complete IGSM application was developed within ServiceNow Studio as a scoped application. The Studio environment contains the custom table, Record Producer, Flow Designer workflow, application roles, ACLs, and other configuration artifacts. The modular structure of the application demonstrates a well-organized enterprise implementation where data, automation, security, and user interface components are maintained within a single application scope.

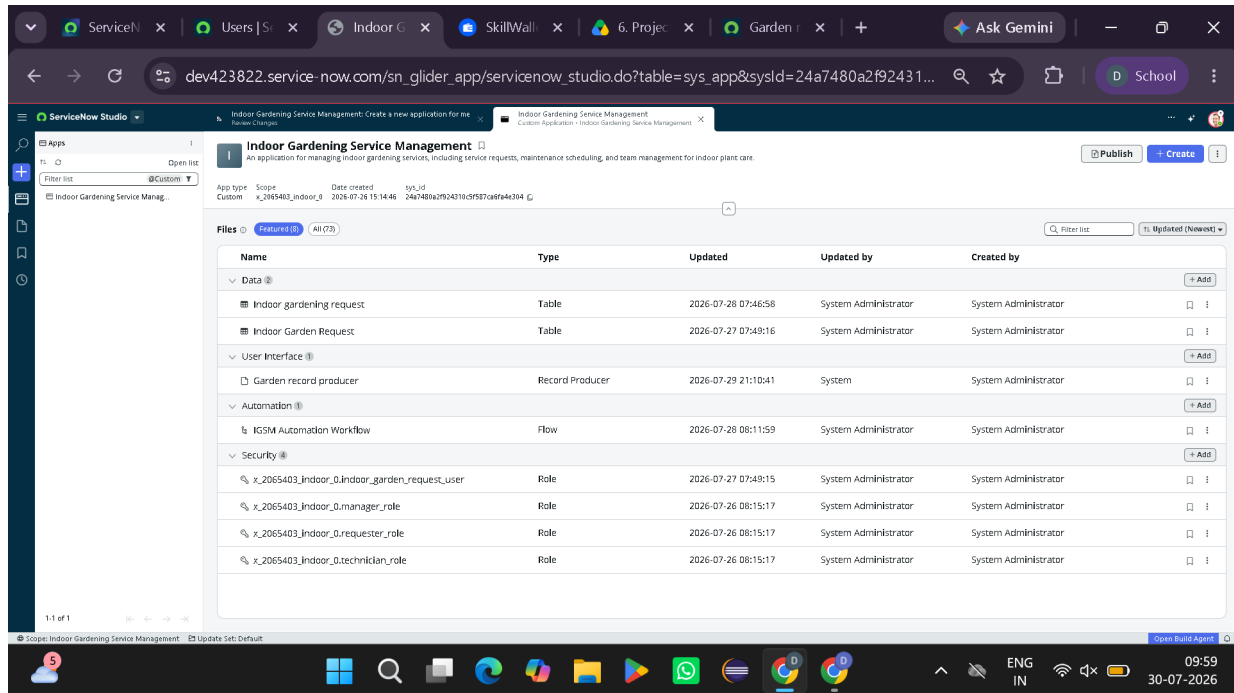


Figure: ServiceNow Studio showing the IGSM application components.

5. Implementation

The implementation was completed through ten milestones: scoped application creation, custom table development, user group configuration, Record Producer implementation, Flow Designer automation, SLA configuration, ACL security, Business Rules, Client Scripts, and end-to-end testing. The custom Indoor Garden Request table stores all gardening requests and extends the ServiceNow Task framework. This provides built-in task management capabilities while allowing the application to maintain gardening-specific information such as plant type, service type, priority, preferred date, and technician assignment.

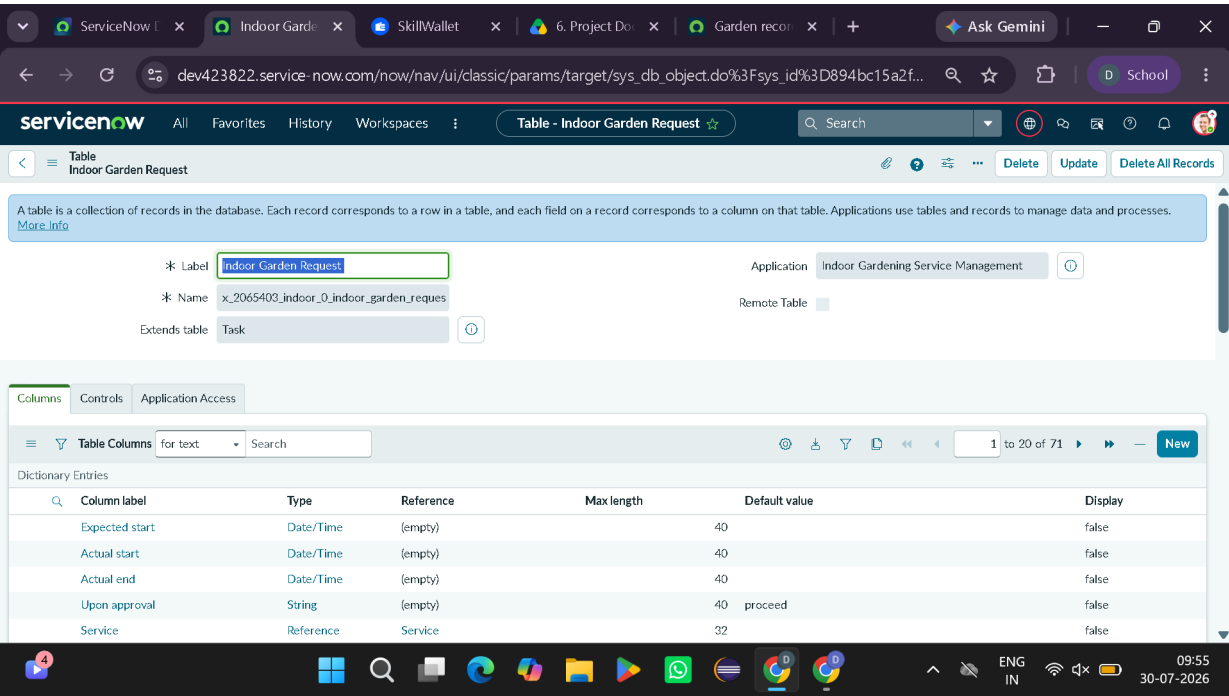
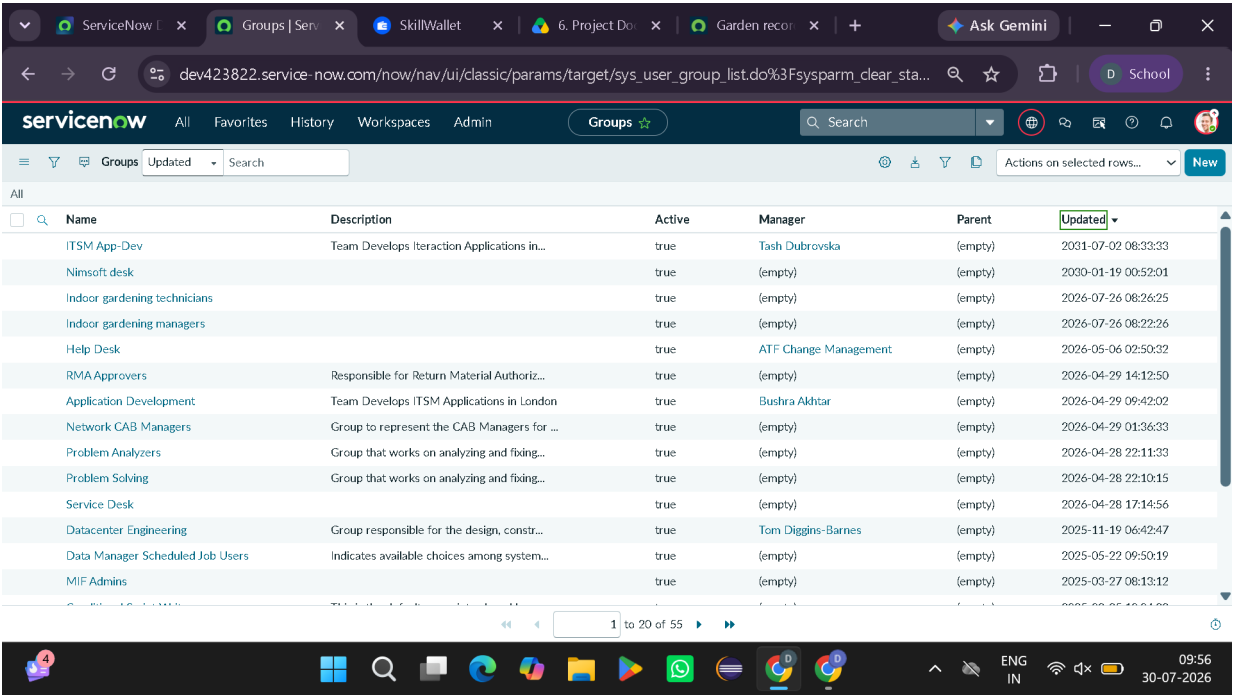


Figure: Indoor Garden Request custom table created in the IGSM scoped application.

5.1 User groups and role-based access

Role-based access control was implemented using dedicated technician and manager groups. The Indoor Gardening Technicians group receives assigned service requests, while the Indoor Gardening Managers group handles approvals and operational supervision. This separation of responsibilities improves security, accountability, and workflow management. The group configuration ensures that assignment rules and approval workflows operate correctly within the application.



The screenshot displays the ServiceNow 'Groups' list page. The table lists various user groups with columns for Name, Description, Active status, Manager, Parent, and an Updated timestamp. The 'Indoor gardening technicians' and 'Indoor gardening managers' groups are highlighted, corresponding to the roles mentioned in the text.

Name	Description	Active	Manager	Parent	Updated
ITSM App-Dev	Team Develops Iteraction Applications in...	true	Tash Dubrovskva	(empty)	2031-07-02 08:33:33
Nimsoft desk		true	(empty)	(empty)	2030-01-19 00:52:01
Indoor gardening technicians		true	(empty)	(empty)	2026-07-26 08:26:25
Indoor gardening managers		true	(empty)	(empty)	2026-07-26 08:22:26
Help Desk		true	ATF Change Management	(empty)	2026-05-06 02:50:32
RMA Approvers	Responsible for Return Material Authoriz...	true	(empty)	(empty)	2026-04-29 14:12:50
Application Development	Team Develops ITSM Applications in London	true	Bushra Akhtar	(empty)	2026-04-29 09:42:02
Network CAB Managers	Group to represent the CAB Managers for ...	true	(empty)	(empty)	2026-04-29 01:36:33
Problem Analyzers	Group that works on analyzing and fixing...	true	(empty)	(empty)	2026-04-28 22:11:33
Problem Solving	Group that works on analyzing and fixing...	true	(empty)	(empty)	2026-04-28 22:10:15
Service Desk		true	(empty)	(empty)	2026-04-28 17:14:56
Datacenter Engineering	Group responsible for the design, constr...	true	Tom Diggins-Barnes	(empty)	2025-11-19 06:42:47
Data Manager Scheduled Job Users	Indicates available choices among system...	true	(empty)	(empty)	2025-05-22 09:50:19
MIF Admins		true	(empty)	(empty)	2025-03-27 08:13:12

Figure: Indoor Gardening Technicians and Managers user groups.

6. Project planning and scheduling

The project was executed across four development sprints. Sprint 1 focused on application setup, data model creation, and user group configuration. Sprint 2 implemented the Record Producer and Flow Designer automation. Sprint 3 covered SLA configuration and ACL security. Sprint 4 implemented Business Rules, Client Scripts, and end-to-end testing. The project completed all planned milestones successfully. The structured milestone approach allowed each major feature to be implemented and validated independently before integrating the complete workflow.

7. Functional and performance testing

Functional testing validated the complete customer request lifecycle. Test cases included Record Producer submission, request creation, approval processing, technician assignment, SLA triggering, ACL validation, and request state transitions. Performance validation focused on Response SLA and Resolution SLA execution. Security testing verified role-based access for requesters, technicians, and managers. The Catalog Client Script was tested by selecting urgent priority and confirming that the description field became mandatory, demonstrating successful client-side validation.

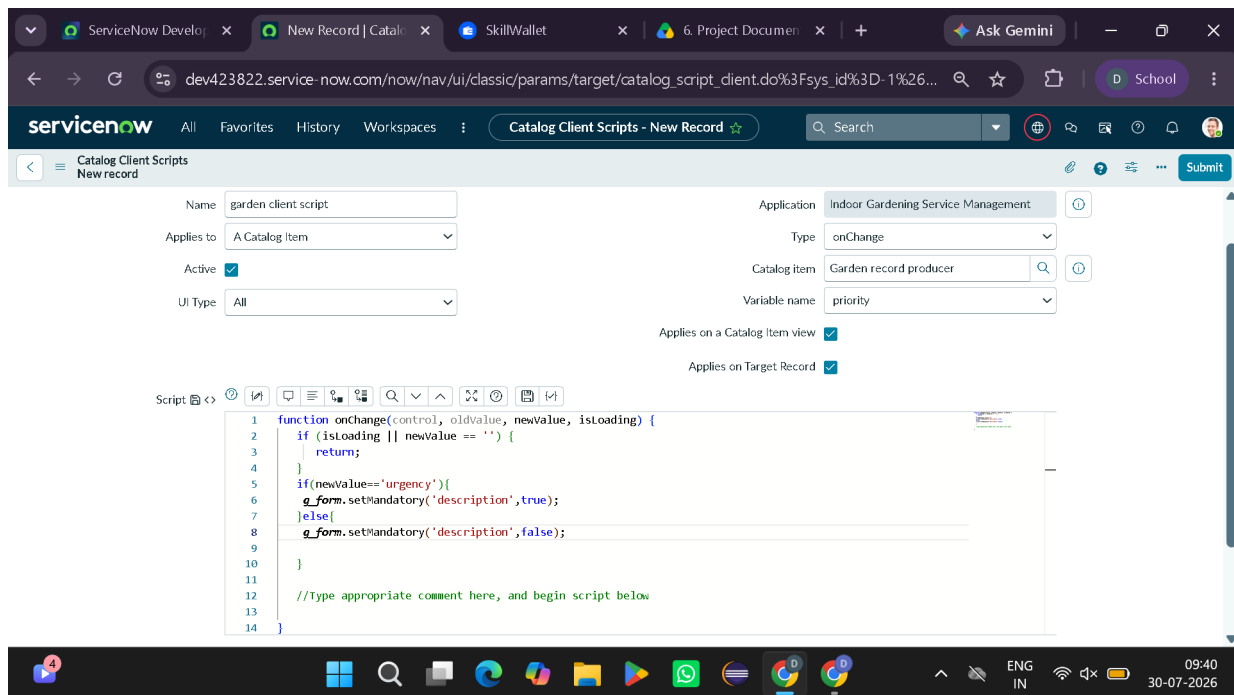
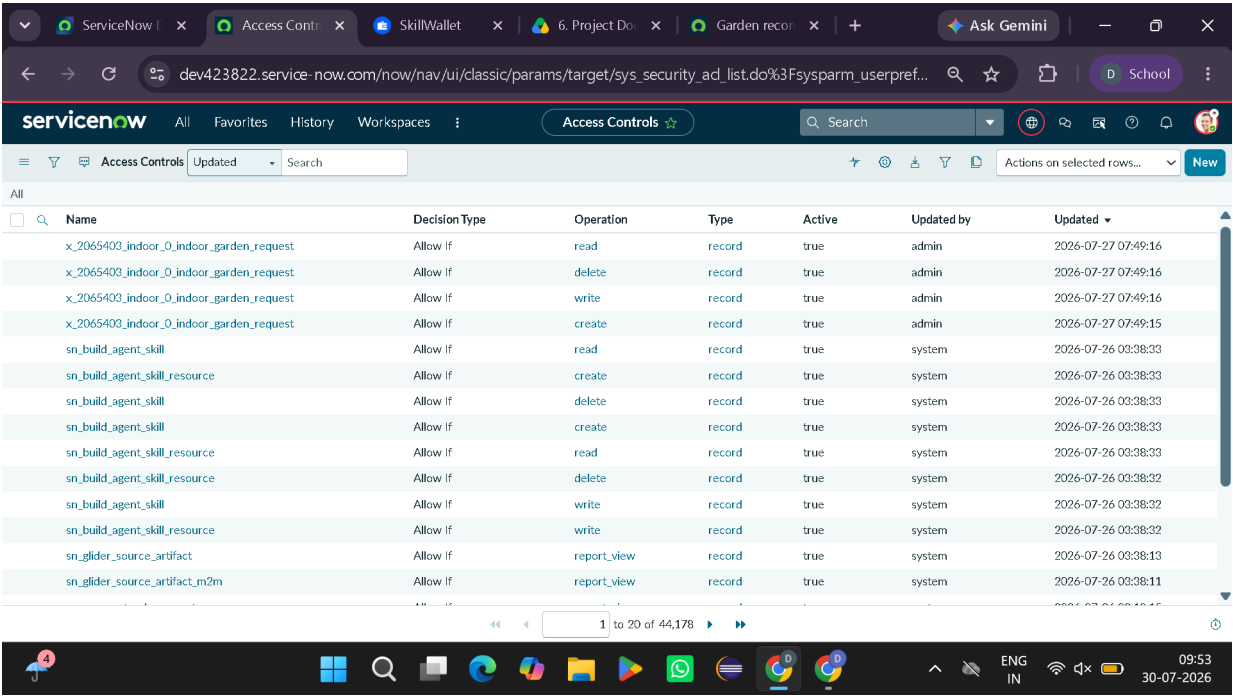


Figure: Catalog Client Script implementing priority-based validation.

7.1 Security testing and ACL validation

Access Control Lists (ACLs) were configured for create, read, write, and delete operations on the Indoor Garden Request table. Multiple user roles were tested to verify that unauthorized access was prevented while operational users retained the required permissions. The ACL configuration demonstrates secure implementation of role-based access control within the IGSM application and validates the security requirements defined during the requirement analysis phase.



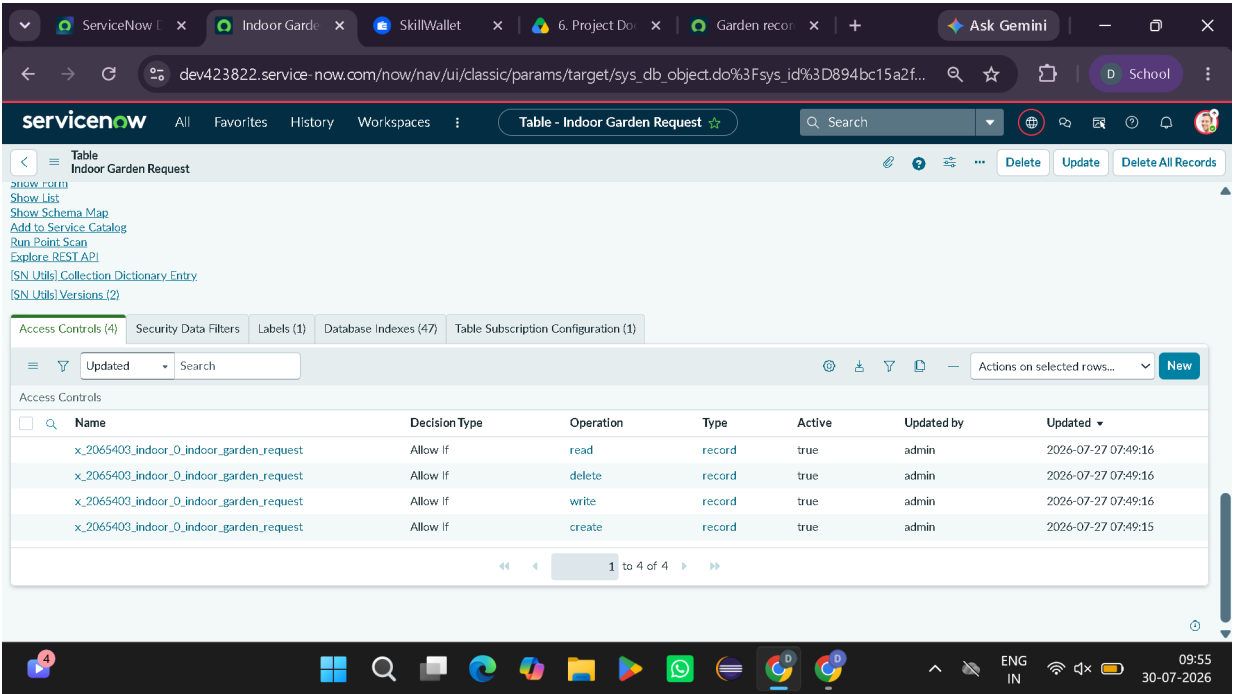
The screenshot displays the ServiceNow 'Access Controls' configuration page. The table lists ACLs for various operations on the 'Indoor Garden Request' table. The columns are: Name, Decision Type, Operation, Type, Active, Updated by, and Updated. The ACLs are configured to allow 'If' for various operations (read, delete, write, create, report_view) on different resources (x_2065403_indoor_0_indoor_garden_request, sn_build_agent_skill_resource, sn_build_agent_skill, sn_glider_source_artifact, sn_glider_source_artifact_m2m).

Name	Decision Type	Operation	Type	Active	Updated by	Updated
x_2065403_indoor_0_indoor_garden_request	Allow If	read	record	true	admin	2026-07-27 07:49:16
x_2065403_indoor_0_indoor_garden_request	Allow If	delete	record	true	admin	2026-07-27 07:49:16
x_2065403_indoor_0_indoor_garden_request	Allow If	write	record	true	admin	2026-07-27 07:49:16
x_2065403_indoor_0_indoor_garden_request	Allow If	create	record	true	admin	2026-07-27 07:49:15
sn_build_agent_skill	Allow If	read	record	true	system	2026-07-26 03:38:33
sn_build_agent_skill_resource	Allow If	create	record	true	system	2026-07-26 03:38:33
sn_build_agent_skill	Allow If	delete	record	true	system	2026-07-26 03:38:33
sn_build_agent_skill	Allow If	create	record	true	system	2026-07-26 03:38:33
sn_build_agent_skill_resource	Allow If	read	record	true	system	2026-07-26 03:38:33
sn_build_agent_skill_resource	Allow If	delete	record	true	system	2026-07-26 03:38:32
sn_build_agent_skill	Allow If	write	record	true	system	2026-07-26 03:38:32
sn_build_agent_skill_resource	Allow If	write	record	true	system	2026-07-26 03:38:32
sn_glider_source_artifact	Allow If	report_view	record	true	system	2026-07-26 03:38:13
sn_glider_source_artifact_m2m	Allow If	report_view	record	true	system	2026-07-26 03:38:11

Figure: ACL configuration for Indoor Garden Request operations.

7.2 Table-level access control

Table-level access controls were verified to ensure that only authorized users could perform CRUD operations on the Indoor Garden Request table. This testing confirmed that security rules were enforced consistently across the application and protected sensitive request information from unauthorized modification.



The screenshot displays the ServiceNow interface for the 'Indoor Garden Request' table. The 'Access Controls' tab is selected, showing a list of four access controls. Each control is named 'x_2065403_indoor_0_indoor_garden_request' and is configured with 'Allow If' as the decision type. The operations are read, delete, write, and create, all with a 'record' type and an 'Active' status of 'true'. The 'Updated by' field for all controls is 'admin', and the 'Updated' timestamp is '2026-07-27 07:49:16'.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
x_2065403_indoor_0_indoor_garden_request	Allow If	read	record	true	admin	2026-07-27 07:49:16
x_2065403_indoor_0_indoor_garden_request	Allow If	delete	record	true	admin	2026-07-27 07:49:16
x_2065403_indoor_0_indoor_garden_request	Allow If	write	record	true	admin	2026-07-27 07:49:16
x_2065403_indoor_0_indoor_garden_request	Allow If	create	record	true	admin	2026-07-27 07:49:15

Figure: Table-level access controls for create, read, write, and delete operations.

8. Results and discussion

The IGSM application successfully automated indoor gardening service management by integrating request creation, approvals, technician assignment, SLA monitoring, and role-based security into a single ServiceNow workflow. The implemented workflow reduced manual coordination, improved request visibility, and enabled structured service delivery. Managers can monitor operational performance, technicians receive automated assignments, and customers can track request progress throughout the service lifecycle.

9. Advantages and limitations

The application provides centralized request management, automated workflows, SLA monitoring, role-based security, and operational transparency. The modular architecture supports future enhancements and integration with additional ServiceNow capabilities. The primary limitation is platform dependency on ServiceNow, and future enhancements could include mobile access, IoT-based plant monitoring, predictive maintenance, customer feedback analytics, and inventory integration.

10. Conclusion

The Indoor Gardening Services Management (IGSM) project successfully achieved its objective of digitizing and automating indoor gardening service operations. The integration of Service Catalog, custom tables, Flow Designer, ACLs, SLAs, and Client Scripts demonstrates a complete enterprise workflow implementation using ServiceNow. The project validates the use of ServiceNow as a flexible workflow automation platform for operational service management and provides a strong foundation for future enterprise enhancements.

11. Future scope

Future development can include mobile application support, IoT-enabled plant health monitoring, predictive maintenance, AI-assisted technician assignment, automated inventory management, and advanced analytics dashboards. These enhancements would further improve service efficiency, reduce maintenance costs, and provide intelligent decision support for indoor gardening operations.

12. Appendix

GitHub Repository: <https://github.com/Dega-Leela-Ramana/Indoor-Gardening-Service-management> The repository contains the ServiceNow scoped application artifacts, custom table configuration, Record Producer implementation, ACLs, SLA configuration, Client Scripts, and project documentation. The screenshots included in this report correspond to the implemented application components available in the project repository.